CMR TECHNICAL CAMPUS

INDUSTRIAL ORIENTED MINI PROJECT - 2026

Detecting and Mitigating Botnet Attacks in Software-Defined Networks Using Deep Learning Techniques

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1. ABSTRACT..

•Software Defined Networking (SDN) is a modern network architecture that separates the **control plane and data plane**.

•It allows centralized network management and flexible configuration.

•However, SDN networks are vulnerable to **cyber attacks like Botnet and DDoS attacks**.

•Botnets are groups of compromised devices controlled by attackers.

•These attacks can overload the SDN controller and disrupt network services.

•This project proposes a **Deep Learning based system** to detect and mitigate botnet attacks.

•The system analyzes network traffic and identifies malicious patterns using machine learning and deep learning models.

2.Existing System

•Traditional intrusion detection systems use **Machine Learning algorithms**.

•Common algorithms used:

- Decision Tree
- Support Vector Machine (SVM)
- Random Forest

•Some research uses **Deep Learning models** like RNN and LSTM.

•These systems detect attacks by analyzing network traffic data.

•Detection depends on feature extraction and dataset quality.

3. Disadvantages

- Difficult to process **large and complex network datasets**.
- Requires **large amount of labeled training data**.
- Incorrect data labeling reduces accuracy.
- High computational time in some algorithms.
- Limited accuracy in detecting advanced botnet attacks.

4. Proposed System

•The proposed system uses **Deep Learning techniques** for botnet detection.

•Network traffic data is collected and preprocessed.

•Important features are selected using **feature weighting and tuning techniques**.

•Multiple algorithms are used for comparison:

- RNN
- CNN
- LSTM
- DNN
- MLP

•The system evaluates model performance using metrics like accuracy, precision, recall, and F1-score.

•Among all models, **CNN provides the best detection performance**.

5. Advantages

- High detection accuracy.
- Faster detection of malicious traffic.
- Efficient feature selection improves performance.
- Handles large datasets effectively.
- Reduces controller overload in SDN networks.
- Improves overall network security.

6. Hardware & Software Requirements

Hardware Requirements

- Processor : Intel i3 / Pentium IV
- RAM : Minimum 4 GB
- Hard Disk : Minimum 20 GB
- Keyboard & Mouse
- Monitor : SVGA

Software Requirements

- Operating System : Windows 7 / Windows 10
- Programming Language : Python
- Framework : Django ORM
- Frontend : HTML, CSS, JavaScript
- Database : MySQL
- Tool : Jupyter Notebook

7. Novelty of the Project

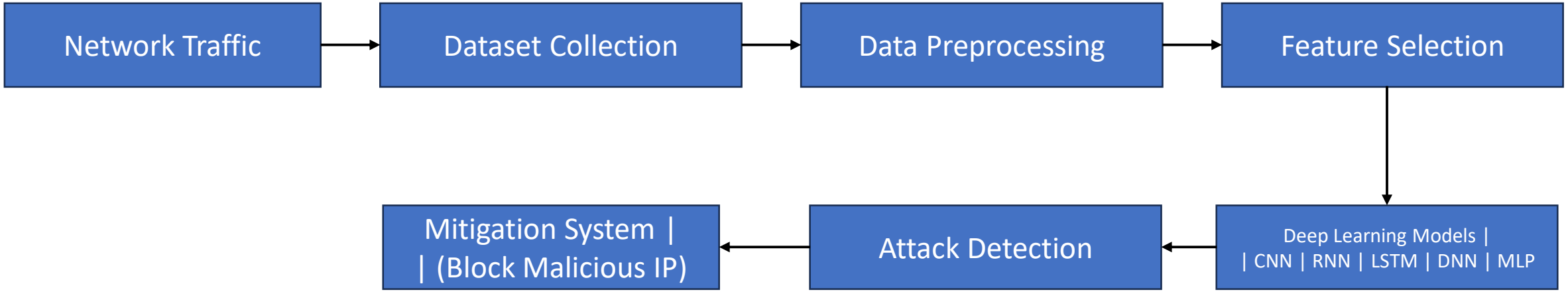
- Uses **Deep Learning models for SDN security**.
- Implements **feature optimization techniques**.
- Compares multiple machine learning and deep learning models.
- Uses **CNN model for high accuracy attack detection**.
- Provides **efficient mitigation mechanism using dynamic flow deletion**.
- Achieves detection accuracy of **around 99%**.

8. Architecture of the System

System Architecture Components:

1. Data Collection Module
2. Data Preprocessing Module
3. Feature Selection Module
4. Machine Learning / Deep Learning Models
5. Attack Detection Module
6. Mitigation Module

Flow:



9. Modules of the System

Main Modules:

1. Dataset Upload

•Upload the network traffic dataset.

2. Dataset Preprocessing

•Cleaning the dataset.

•Converting non-numeric values to numeric values.

•Handling imbalance using SMOTE.

3. Model Training

•Training machine learning and deep learning algorithms.

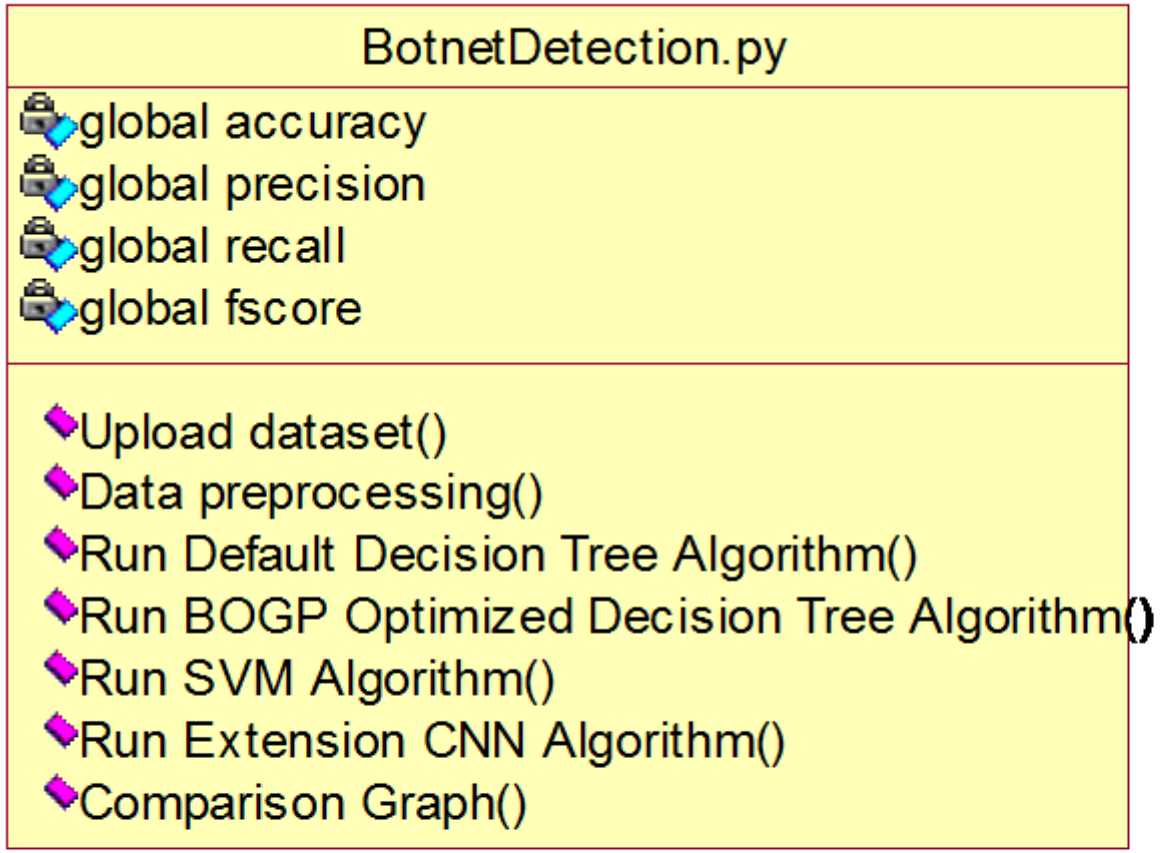
4. Attack Detection

•Classifying traffic as **Normal or Attack**.

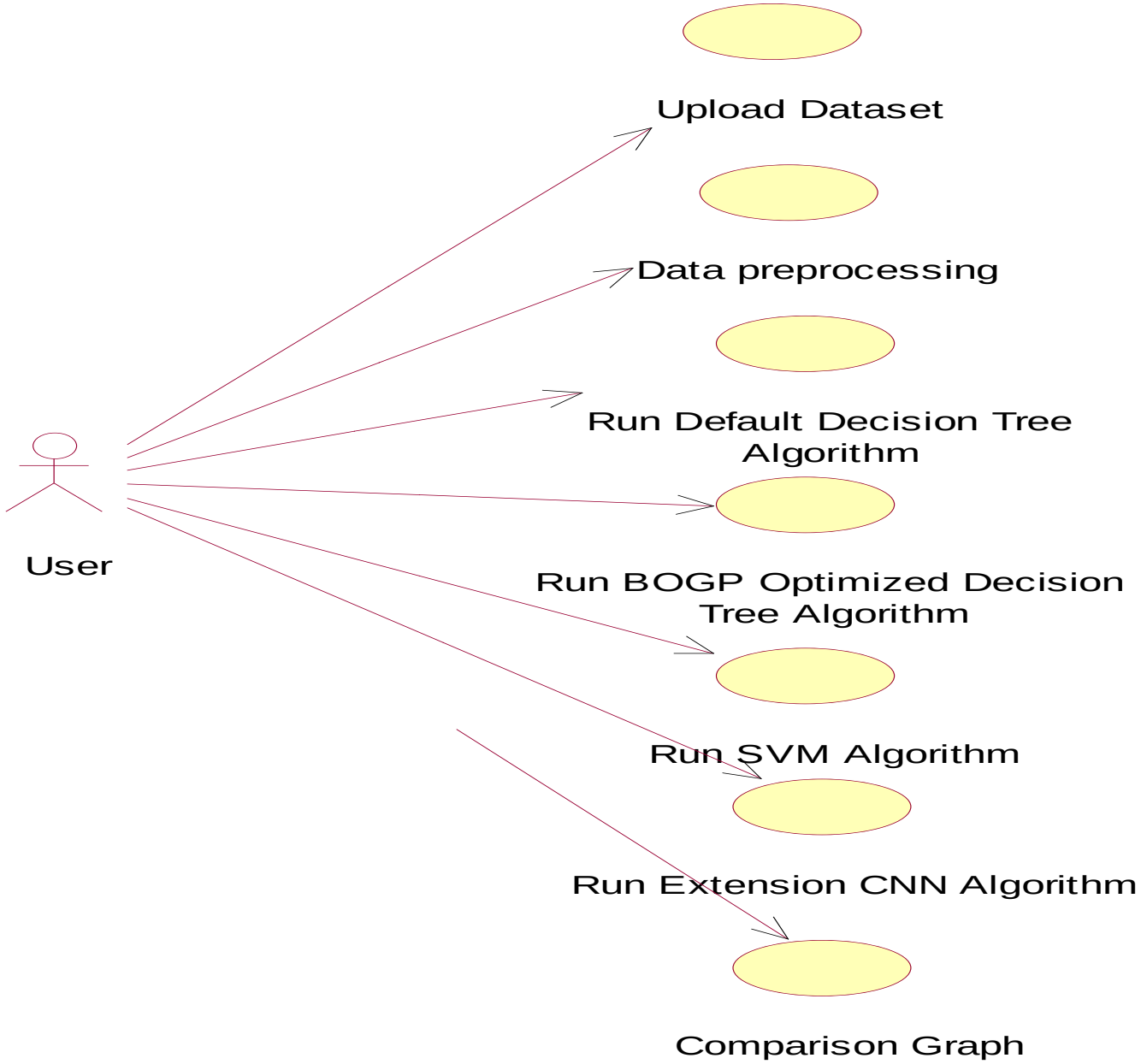
5. Performance Analysis

•Comparing algorithms using accuracy, precision, recall and F1 score.

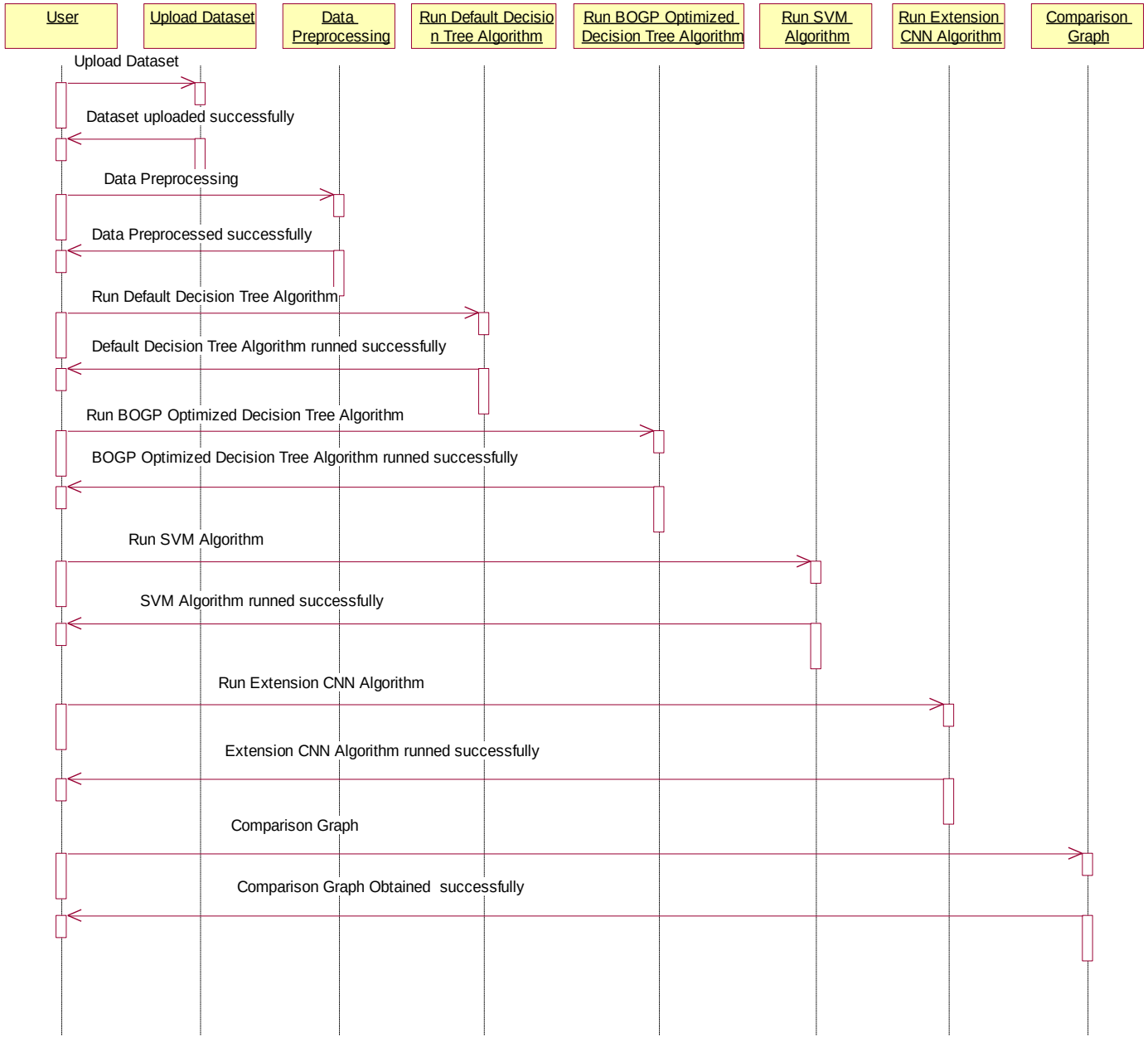
10.CLASS DIAGRAM:



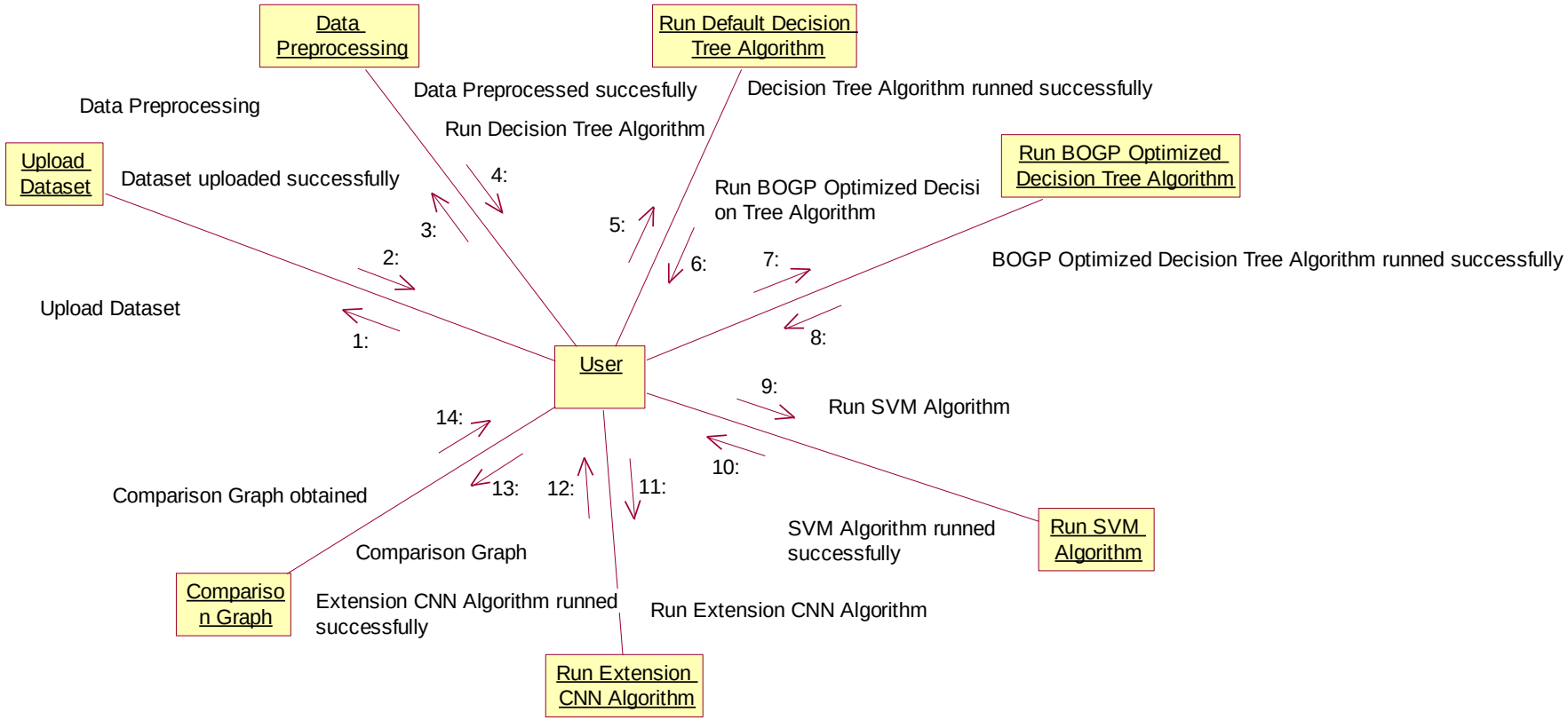
11.USECASE DIAGRAM:



12.SEQUENCE DIAGRAM:



13.COLLABORATION DIAGRAM:



14.Sample Code:

```
import pandas as pd
import numpy as np
from bayes_opt import BayesianOptimization #importing byaseian optimization class
import matplotlib. pyplot as plt #use to visualize dataset values
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import precision_score
from sklearn.metrics import recall_score
from sklearn.metrics import f1_score
from sklearn.metrics import f1_score
from sklearn.metrics import accuracy_score
from sklearn.metrics import confusion_matrix
from sklearn.metrics import roc_curve
from sklearn.metrics import roc_auc_score
from sklearn import metrics
from sklearn.preprocessing import LabelEncoder
from sklearn import svm
import os
from keras.utils.np_utils import to_categorical
from keras.layers import  MaxPooling2D
from sklearn import svm
import os
from keras.utils.np_utils import to_categorical
from keras.layers import  MaxPooling2D
from keras.layers import Dense, Dropout, Activation, Flatten, GlobalAveragePooling2D, BatchNormalization
from keras.layers import Convolution2D
from keras.models import Sequential
from keras.callbacks import ModelCheckpoint
import pickle
dataset = pd.read_csv("Dataset/UNSW_2018_IoT_Botnet_Full5pc_4.csv", low_memory=False)
dataset
labels, count = np.unique(dataset['attack'].ravel(), return_counts = True)
print("Normal Records : "+str(count[0]))
print("Attack Records : "+str(count[1]))
height = count
bars = labels
y_pos = np.arange(len(bars))
plt.figure(figsize = (4, 3))
plt.bar(y_pos, height)
```

```
plt.xticks(y_pos, bars)
plt.xlabel("Dataset Class Label Graph")
plt.ylabel("Count")
plt.xticks(rotation=90)
plt.show()
dataset.isnull().sum()
dataset.groupby("proto").size().plot.pie(autopct='%.0f%%',
figsize=(6, 6))
plt.title("Different Protocol Graph")
plt.xlabel("Protocol Name")
plt.ylabel("Usage %")
plt.show()
data = dataset[['proto', 'attack']]
plt.figure(figsize=(6,4))
sns.boxplot(data=data, x='proto', y='attack', palette='rainbow')
plt.title("Different Protocols Used by Attacks")
plt.show()
dataset.drop(['pkSeqID', 'category', 'subcategory'], axis =
1,inplace=True)
#using label encoder to convert non-numeric values to numeric
values
encoder1 = LabelEncoder()
encoder2 = LabelEncoder()
encoder3 = LabelEncoder()
encoder4 = LabelEncoder()
encoder5 = LabelEncoder()
encoder6 = LabelEncoder()
encoder7 = LabelEncoder()
dataset['flgs'] =
pd.Series(encoder1.fit_transform(dataset['flgs'].astype(str)))#encode
all str columns to numeric
dataset['proto'] =
pd.Series(encoder2.fit_transform(dataset['proto'].astype(str)))#encod
e all str columns to numeric
dataset['saddr'] =
pd.Series(encoder3.fit_transform(dataset['saddr'].astype(str)))#encod
e all str columns to numeric
dataset['sport'] =
pd.Series(encoder4.fit_transform(dataset['sport'].astype(str)))#encod
e all str columns to numeric
dataset['daddr'] =
pd.Series(encoder5.fit_transform(dataset['daddr'].astype(str)))#encod
e all str columns to numeric
dataset['dport'] =
pd.Series(encoder6.fit_transform(dataset['dport'].astype(str)))#encod
e all str columns to numeric
```

```
dataset['state'] = pd.Series(encoder7.fit_transform(dataset['state'].astype(str)))#encode all str columns to numeric
Dataset
pca = PCA(n_components=2)
temp = dataset.values
X = temp[:,0:temp.shape[1]-1]
Y = temp[:,temp.shape[1]-1]
X = pca.fit_transform(X)
plt.figure(figsize=(6,4))
plt.scatter(X[:, 0], X[:, 1], c=Y)
plt.xlabel("First Principal Component")
plt.ylabel("Second Principal Component")
plt.title("Dataset Imbalance Graph")
plt.show()
Y = dataset['attack'].ravel()#represents attack or normal
data = dataset.values
X = data[:,0:dataset.shape[1]-1]
indices = np.arange(X.shape[0])
np.random.shuffle(indices)#shuffling dataset values
X = X[indices]
Y = Y[indices]
thr = 50
#normalizing dataset values using min max scaler
scaler = MinMaxScaler(feature_range = (0, 1))
X = scaler.fit_transform(X)#normalize train features
print("Normalize Training Features")
print(X)
#now apply smote algorithm on imbalance dataset for balancing
smote = SMOTE()
X, Y = smote.fit_resample(X, Y)
#dataset class labels after applying smote
labels, count = np.unique(Y, return_counts = True)
print("Normal Records After Smote : "+str(count[0]))
print("Attack Records After Smote: "+str(count[1]))
height = count
bars = ['Normal', 'Attack']
y_pos = np.arange(len(bars))
plt.figure(figsize = (4, 3))
plt.bar(y_pos, height)
plt.xticks(y_pos, bars)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size = 0.2)
print("Total records found in dataset = "+str(X.shape[0]))
print("Total features found in dataset= "+str(X.shape[1]))
print("80% dataset for training : "+str(X_train.shape[0]))
print("20% dataset for testing  : "+str(X_test.shape[0]))
#define global variables to save accuracy and other metrics
accuracy = []
precision = []
recall = []
fscore = []
#function to calculate all metrics
def calculateMetrics(algorithm, testY, predict):
    p = precision_score(testY, predict,average='macro') * 100
    r = recall_score(testY, predict,average='macro') * 100
    f = f1_score(testY, predict,average='macro') * 100
    a = accuracy_score(testY,predict)*100
    accuracy.append(a)
    precision.append(p)
    recall.append(r)
    fscore.append(f)
    print(algorithm+" Accuracy  : "+str(a))
    print(algorithm+" Precision : "+str(p))
    print(algorithm+" Recall    : "+str(r))
    print(algorithm+" FSCORE    : "+str(f))
    labels = ['Normal', 'Attack']
    conf_matrix = confusion_matrix(testY, predict)
    fig, axs = plt.subplots(1,2,figsize=(10, 3))
    ax = sns.heatmap(conf_matrix, xticklabels = labels, yticklabels = labels, annot = True, cmap="viridis" ,fmt ="g", ax=axs[0]);
    ax.set_ylim([0,len(labels)])
    axs[0].set_title(algorithm+" Confusion matrix")

    random_probs = [0 for i in range(len(testY))]
```

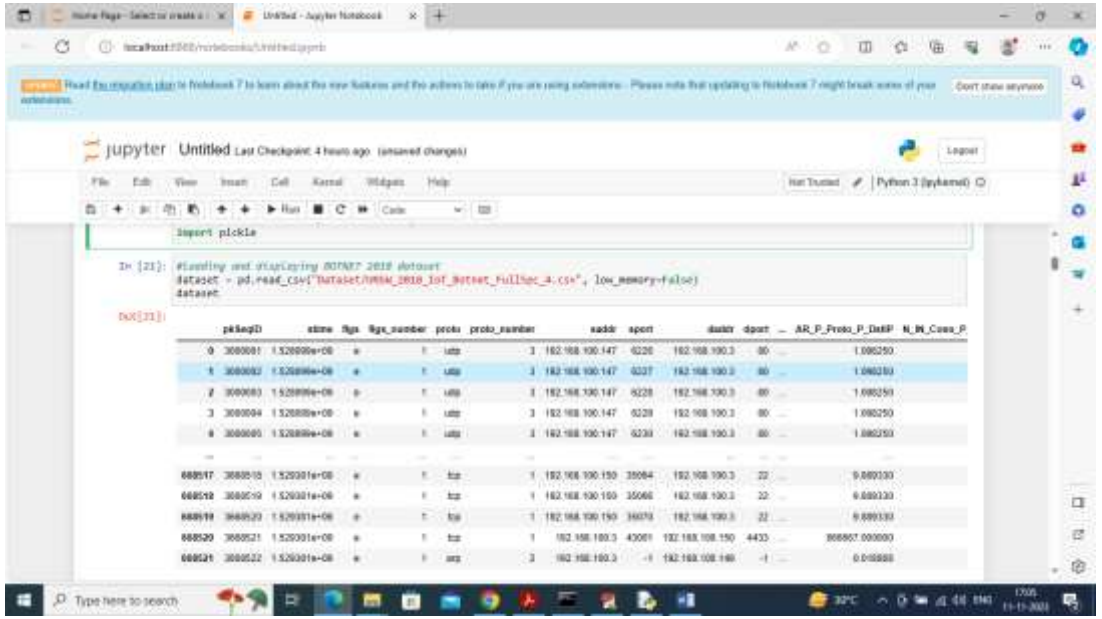
```
p_fpr, p_tpr, _ = roc_curve(testY, random_probs, pos_label=1)
    plt.plot(p_fpr, p_tpr, linestyle='--', color='orange',label="True classes")
    ns_fpr, ns_tpr, _ = roc_curve(testY, predict, pos_label=1)
    axs[1].plot(ns_fpr, ns_tpr, linestyle='--', label='Predicted Classes')
    axs[1].set_title(algorithm+" ROC AUC Curve")
    axs[1].set_xlabel('False Positive Rate')
    axs[1].set_ylabel('True Positive rate')
    plt.show()
#training and evaluating performance of default decision tree algorithm
dt_cls = DecisionTreeClassifier(max_depth=1)
dt_cls.fit(X_train, y_train)#train algorithm using training features and target value
predict =dt_cls.predict(X_test)#perform prediction on test data
#call this function with true and predicted values to calculate accuracy and other metrics
calculateMetrics("Default Decision Tree Algorithm", y_test, predict)
def gaussianProcess(max_depth, min_samples_split, max_features):
    params_dt = {}
    params_dt['max_depth'] = max_depth
    params_dt['min_samples_split'] = min_samples_split
    params_dt['max_features'] = max_features
    scores = cross_val_score(DecisionTreeClassifier(random_state=123, **params_dt),
                             X_train, y_train, cv=5).mean()
    score = scores.mean()
    return score
params_dt = {'max_depth':(5, 10), 'min_samples_split':(0.1, 0.9), 'max_features':(0.1, 0.9)}
dt_bo = BayesianOptimization(gaussianProcess, params_dt, random_state=111)
dt_bo.maximize(init_points=5, n_iter=2)
params_dt = dt_bo.max['params']
```

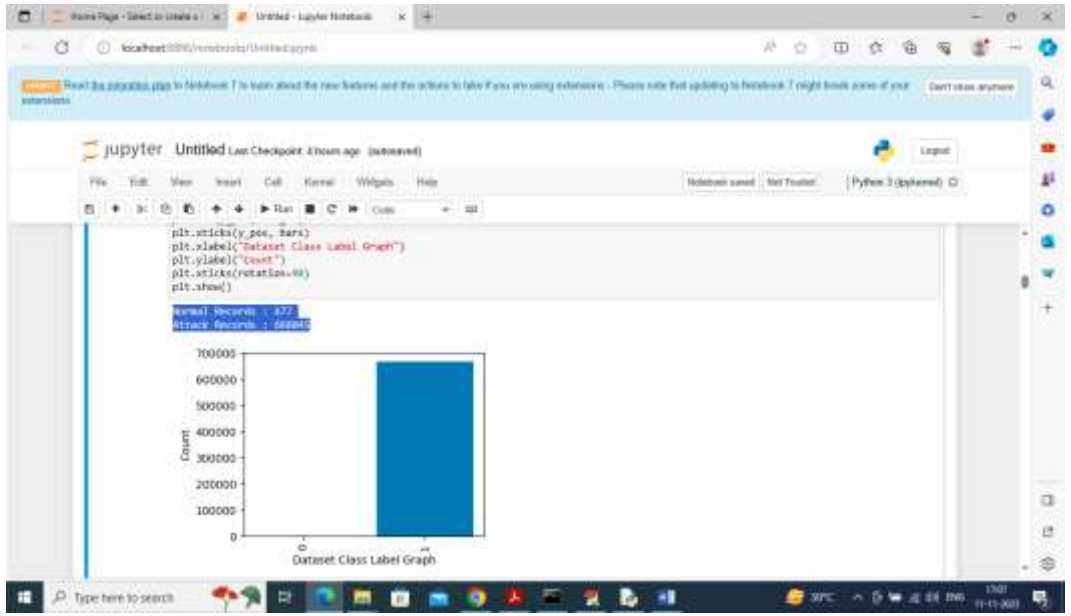
```
print("Best Optimized Parameters Selected by BOGP = "+str(params_dt))
#training and evaluating performance of BOGP optimizied decision tree algorithm
bogp_dt_cls = DecisionTreeClassifier(max_depth=params_dt['max_depth'], max_features=params_dt['max_features'],

min_samples_split=params_dt['min_samples_split'])
bogp_dt_cls.fit(X_train, y_train)#train algorithm using training features and target value
predict = bogp_dt_cls.predict(X_test)#perform prediction on test data
#call this function with true and predicted values to calculate accuracy and other metrics
calculateMetrics("BOGP Optimized Decision Tree", y_test, predict)
#training and evaluating performance of SVM algorithm
svm_cls = svm.SVC()
svm_cls.fit(X_train[0:thr], y_train[0:thr])#train algorithm using training features and target value
predict = svm_cls.predict(X_test) #perform prediction on test data
#call this function with true and predicted values to calculate accuracy and other metrics
calculateMetrics("SVM Algorithm", y_test, predict)
X_train1 = np.reshape(X_train, (X_train.shape[0], X_train.shape[1], 1, 1))
X_test1 = np.reshape(X_test, (X_test.shape[0], X_test.shape[1], 1, 1))
y_train1 = to_categorical(y_train)
y_test1 = to_categorical(y_test)
#creating deep learning cnn model object
cnn_model = Sequential()
#defining CNN layer wwith 32 neurons of size 1 X 1 to filter dataset features 32 times
cnn_model.add(Convolution2D(32, (1 , 1), input_shape = (X_train1.shape[1], X_train1.shape[2], X_train1.shape[3]), activation = 'relu'))
#defining maxpool layet to collect relevant filtered features from previous CNN layer
cnn_model.add(MaxPooling2D(pool_size = (1, 1)))
#creating another CNN layer with 16 neurons to optimzed features 16 times
```

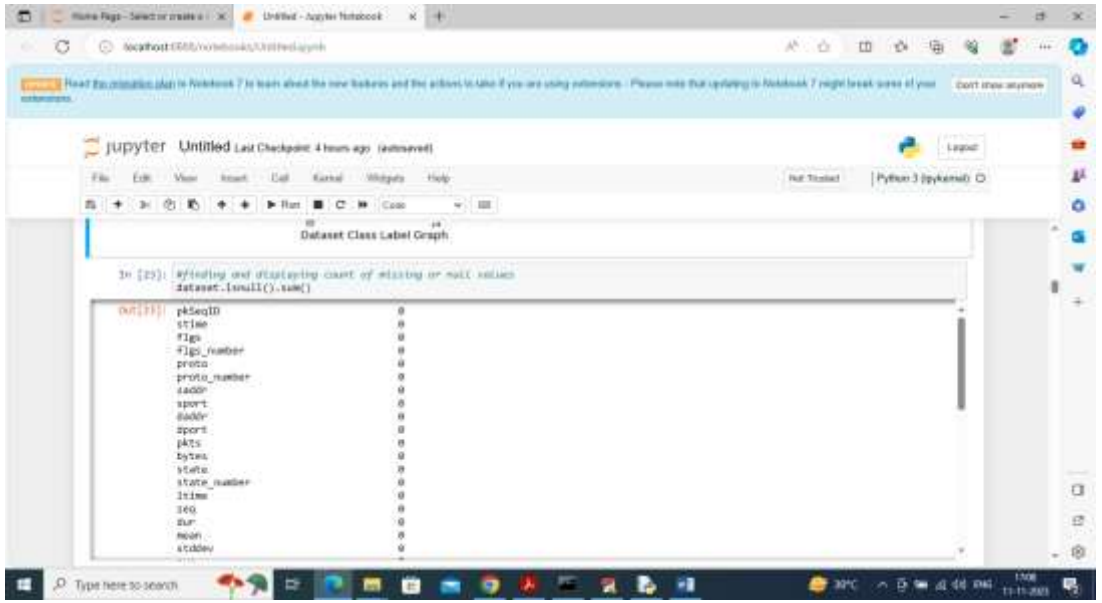
```
cnn_model.add(Convolution2D(16, (1, 1), activation = 'relu'))
cnn_model.add(MaxPooling2D(pool_size = (1, 1)))
#convert multidimension features to single flatten size
cnn_model.add(Flatten())
#define output prediction layer
cnn_model.add(Dense(units = 256, activation = 'relu'))
cnn_model.add(Dense(units = y_train1.shape[1], activation =
'softmax'))
#compile, train and load CNN model
cnn_model.compile(optimizer = 'adam', loss =
'categorical_crossentropy', metrics = ['accuracy'])
if os.path.exists("model/cnn_weights.hdf5") == False:
    model_check_point =
ModelCheckpoint(filepath='model/cnn_weights.hdf5', verbose =
1, save_best_only = True)
    hist = cnn_model.fit(X_train1, y_train1, batch_size = 32,
epochs = 5, validation_data=(X_test1, y_test1),
callbacks=[model_check_point], verbose=1)
    f = open('model/cnn_history.pckl', 'wb')
pickle.dump(hist.history, f)
    f.close()
else:
    cnn_model.load_weights("model/cnn_weights.hdf5")
predict = cnn_model.predict(X_test1)
predict = np.argmax(predict, axis=1)
y_test1 = np.argmax(y_test1, axis=1)
#call this function to calculate accuracy and other metrics
calculateMetrics("Extension CNN", y_test1, predict)
#comparison graph between all algorithms
df = pd.DataFrame([['Default Decision
Tree','Accuracy',accuracy[0]],['Default Decision
Tree','Precision',precision[0]],['Default Decision
Tree','Recall',recall[0]],['Default Decision Tree','FSCORE',fscore[0]],
                ['Optimized Decision
Tree','Accuracy',accuracy[1]],['Optimized Decision
Tree','Precision',precision[1]],['Optimized Decision
Tree','Recall',recall[1]],['Optimized Decision Tree','FSCORE',fscore[1]],
['SVM','Accuracy',accuracy[2]],['SVM','Precision',precision[2]],['SVM','Re
call',recall[2]],['SVM','FSCORE',fscore[2]],
                ['Extension CNN','Accuracy',accuracy[3]],['Extension
CNN','Precision',precision[3]],['Extension
CNN','Recall',recall[3]],['Extension CNN','FSCORE',fscore[3]],
                ],columns=['Parameters','Algorithms','Value'])
df.pivot("Parameters", "Algorithms", "Value").plot(kind='bar', figsize=(6,
3))
plt.title("All Algorithms Performance Graph")
```

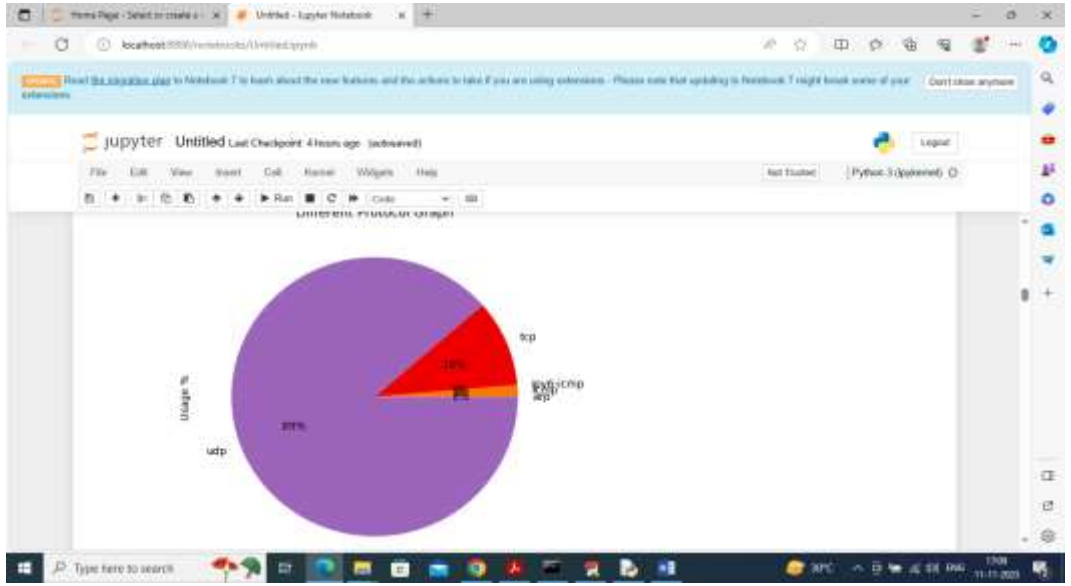
```
plt.show()
#display all algorithm performnace
algorithms = ['Default Decision Tree', 'Optimized Decision Tree', 'SVM', 'Extension CNN']
data = []
for i in range(len(accuracy)):
    data.append([algorithms[i], accuracy[i], precision[i], recall[i], fscore[i]])
data = pd.DataFrame(data, columns=['Algorithm Name', 'Accuracy', 'Precision', 'Recall', 'FSCORE'])
data
```

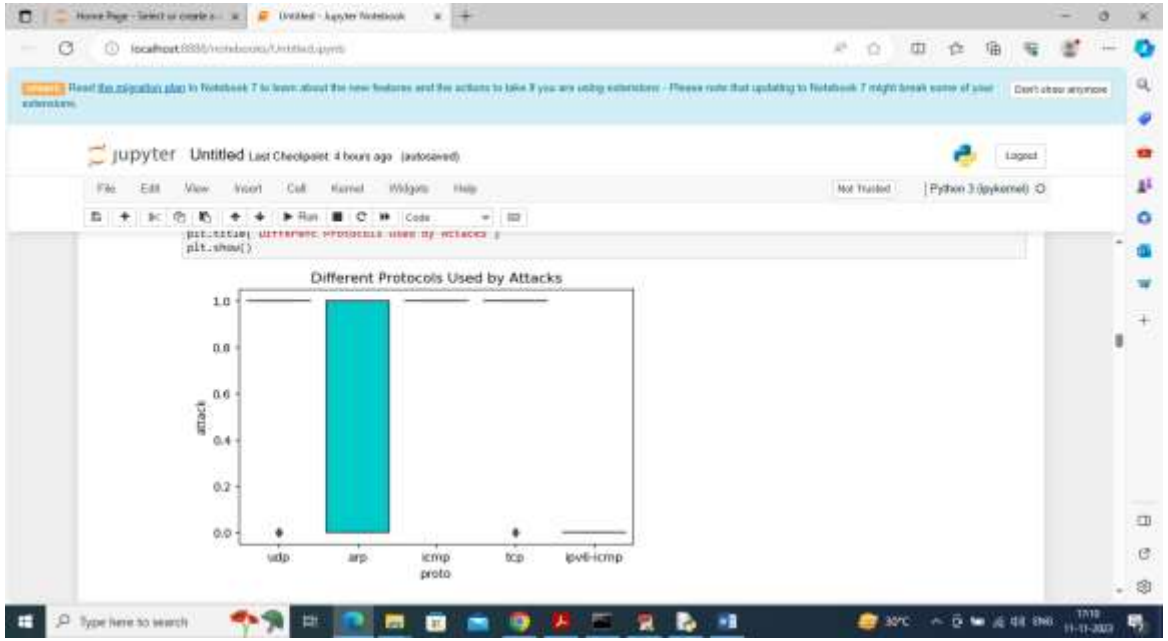
15.SAMPLE OUTPUTS

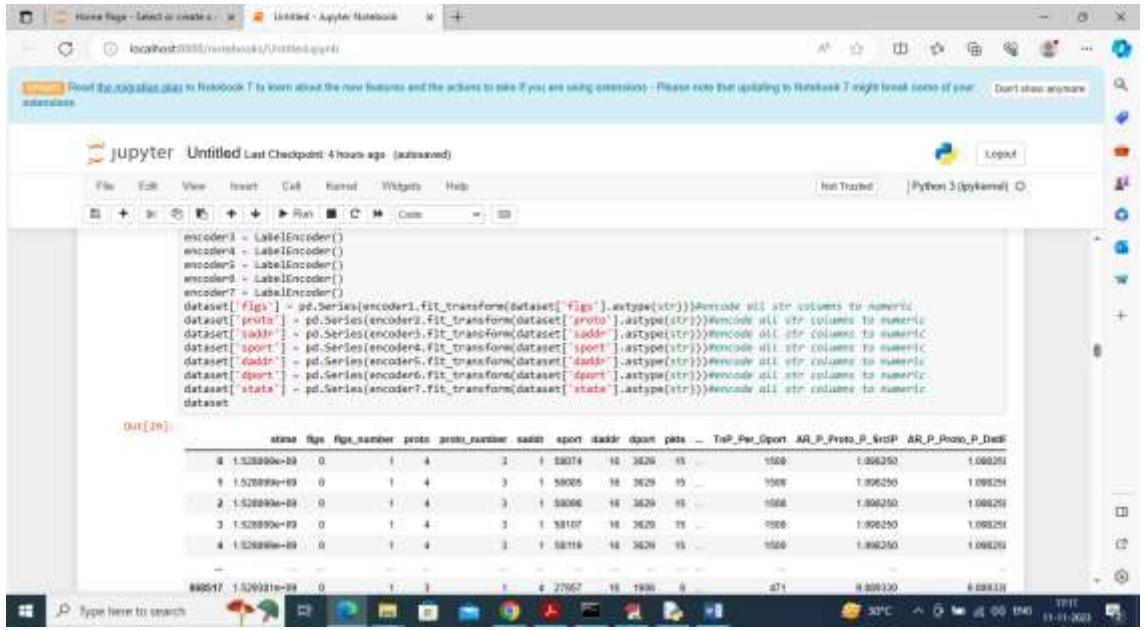


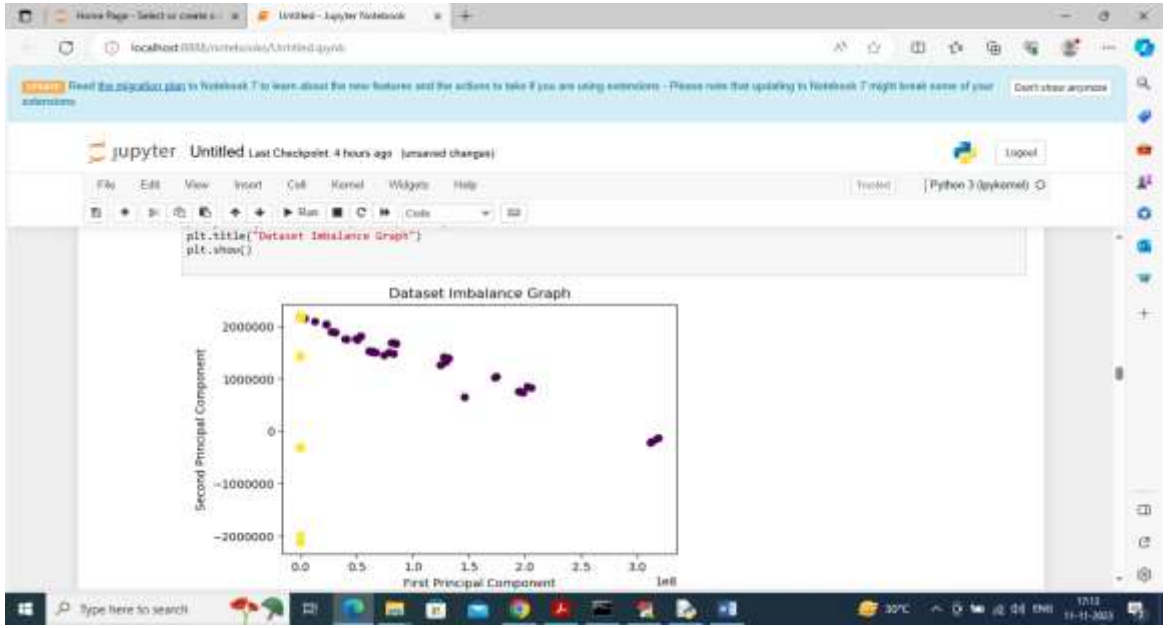


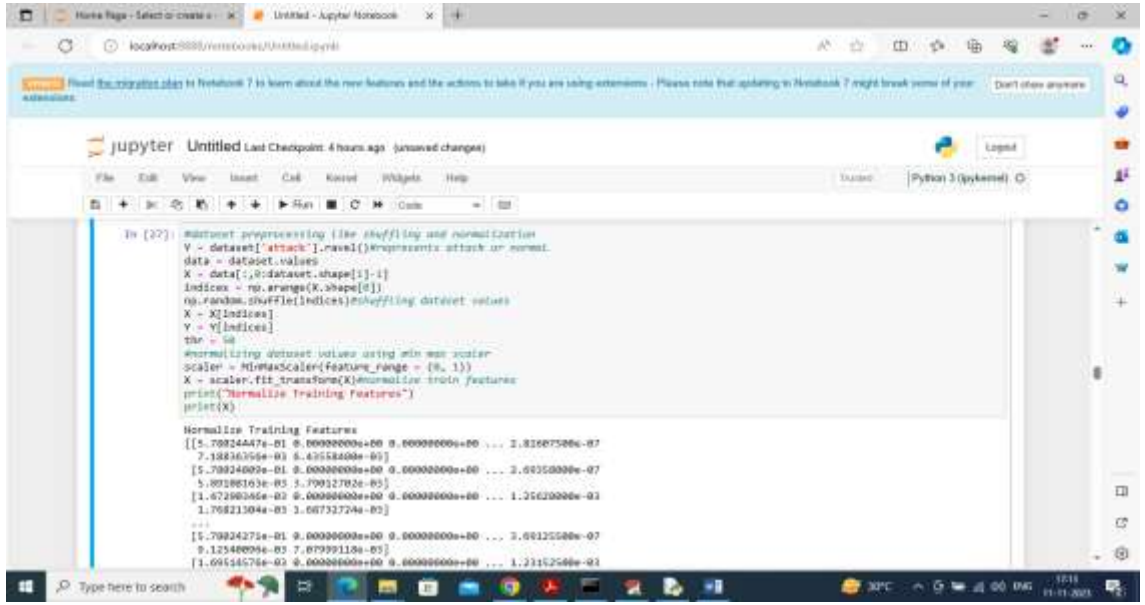





jupyter Untitled Last Checkpoint: 4 hours ago (autosaved)
plt.title('Different Protocols Used by Attacks')
plt.show()
Different Protocols Used by Attacks
attack
proto
udp
arp
icmp
tcp
ipv6-icmp

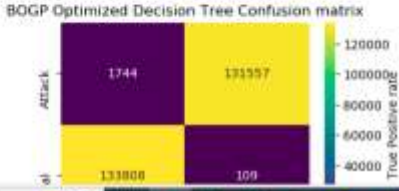

jupyter Untitled Last Checkpoint: 4 hours ago (autosaved)
encoder1 = LabelEncoder()
encoder2 = LabelEncoder()
encoder3 = LabelEncoder()
encoder4 = LabelEncoder()
encoder5 = LabelEncoder()
encoder6 = LabelEncoder()
encoder7 = LabelEncoder()
dataset['flgs'] = pd.Series(encoder1.fit_transform(dataset['flgs'].astype(str)))#encode all str columns to numeric
dataset['proto'] = pd.Series(encoder2.fit_transform(dataset['proto'].astype(str)))#encode all str columns to numeric
dataset['saddr'] = pd.Series(encoder3.fit_transform(dataset['saddr'].astype(str)))#encode all str columns to numeric
dataset['sport'] = pd.Series(encoder4.fit_transform(dataset['sport'].astype(str)))#encode all str columns to numeric
dataset['daddr'] = pd.Series(encoder5.fit_transform(dataset['daddr'].astype(str)))#encode all str columns to numeric
dataset['dport'] = pd.Series(encoder6.fit_transform(dataset['dport'].astype(str)))#encode all str columns to numeric
dataset['state'] = pd.Series(encoder7.fit_transform(dataset['state'].astype(str)))#encode all str columns to numeric
dataset
Out[26]:


jupyter Untitled Last Checkpoint: 4 hours ago (unsaved changes)
plt.title("Dataset Imbalance Graph")
plt.show()
Dataset Imbalance Graph
Second Principal Component
First Principal Component

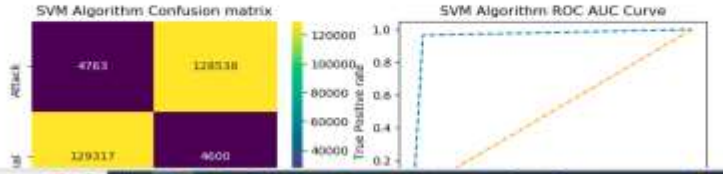

jupyter Untitled Last Checkpoint: 4 hours ago (unsaved changes)
In [27]:
#dataset preprocessing like shuffling and normalization
Y = dataset['attack'].ravel()#represents attack or normal
data = dataset.values
X = data[:,0:dataset.shape[1]-1]
indices = np.arange(X.shape[0])
np.random.shuffle(indices)#shuffling dataset values
X = X[indices]
Y = Y[indices]
thr = 50
#normalizing dataset values using min max scaler
scaler = MinMaxScaler(feature_range = (0, 1))
X = scaler.fit_transform(X)#normalize train features
print("Normalize Training Features")
print(X)
Normalize Training Features

```
In [34]: #giving different parameters as input to bayesian optimization and by using this parameters will process using BOGP to
         #select best parameters
         params_dt = {'max_depth':(5, 10), 'min_samples_split':(0.1, 0.9), 'max_features':(0.1, 0.9)}
         dt_bo = BayesianOptimization(gaussianProcess, params_dt, random_state=111)
         dt_bo.maximize(init_points=5, n_iter=2)
         params_dt = dt_bo.max['params']
         print("Best Optimized Parameters Selected by BOGP = "+str(params_dt))

| iter    | target  | max_depth | max_fe... | min_sa... |
---------------------------------------------------------------
| 1       | 0.9821  | 8.061     | 0.2353    | 0.4488    |
| 2       | 0.9943  | 8.946     | 0.3383    | 0.2193    |
| 3       | 0.9928  | 5.112     | 0.4362    | 0.2909    |
| 4       | 0.9905  | 6.988     | 0.8026    | 0.3982    |
| 5       | 0.9921  | 5.406     | 0.6357    | 0.597     |
| 6       | 0.9995  | 6.584     | 0.9       | 0.1101    |
| 7       | 0.9988  | 6.48      | 0.2044    | 0.1156    |
===============================================================
Best Optimized Parameters Selected by BOGP = {'max_depth': 8.946394594439461, 'max_features': 0.8935694971704703, 'min_samples_s
plit': 0.2909181931201824}

In [35]: #training and evaluating performance of BOGP optimized decision tree algorithm
```

```
In [35]: #training and evaluating performance of BOGP optimized decision tree algorithm
         bogp_dt_cls = DecisionTreeClassifier(max_depth=params_dt['max_depth'], max_features=params_dt['max_features'],
                                              min_samples_split=params_dt['min_samples_split'])
         bogp_dt_cls.fit(X_train, y_train)#train algorithm using training features and target value
         predict = bogp_dt_cls.predict(X_test)#perform prediction on test data
         #call this function with true and predicted values to calculate accuracy and other metrics
         calculateMetrics("BOGP Optimized Decision Tree", y_test, predict)

         BOGP Optimized Decision Tree Accuracy  : 99.30655868990861
         BOGP Optimized Decision Tree Precision : 99.31531182358304
         BOGP Optimized Decision Tree Recall    : 99.30514451699025
         BOGP Optimized Decision Tree FSCORE    : 99.30650947914538
```



```
In [36]: #training and evaluating performance of SVM algorithm
         svm_cls = svm.SVC()
         svm_clx.fit(X_train[0:thr], y_train[0:thr])#train algorithm using training features and target value
         predict = svm_cls.predict(X_test) #perform prediction on test data
         #call this function with true and predicted values to calculate accuracy and other metrics
         calculateMetrics("SVM Algorithm", y_test, predict)

         SVM Algorithm Accuracy  : 96.49611927340224
         SVM Algorithm Precision : 96.49629076627755
         SVM Algorithm Recall    : 96.49500003587244
         SVM Algorithm FSCORE    : 96.49600949533538
```




```
In [37]: #training CNN deep learning algorithm to predict factory maintenance
         #converting dataset shape for CNN compatible format as 4 dimension array
         X_train1 = np.reshape(X_train, (X_train.shape[0], X_train.shape[1], 1, 1))
         X_test1 = np.reshape(X_test, (X_test.shape[0], X_test.shape[1], 1, 1))
         y_train1 = to_categorical(y_train)
         y_test1 = to_categorical(y_test)
         #creating deep learning cnn model object
         cnn_model = Sequential()
         #defining CNN layer with 32 neurons of size 1 X 1 to filter dataset features 32 times
         cnn_model.add(Convolution2D(32, (1 , 1), input_shape = (X_train1.shape[1], X_train1.shape[2], X_train1.shape[3]), activation = '
         #defining maxpool layer to collect relevant filtered features from previous CNN layer
         cnn_model.add(MaxPooling2D(pool_size = (1, 1)))
         #creating another CNN layer with 20 neurons to optimized features 16 times
         cnn_model.add(Convolution2D(16, (1, 1), activation = 'relu'))
         #max layer to collect relevant features
         cnn_model.add(MaxPooling2D(pool_size = (1, 1)))
         #convert multidimension features to single flatten size
         cnn_model.add(Flatten())
         #define output prediction layer
         cnn_model.add(Dense(units = 256, activation = 'relu'))
         cnn_model.add(Dense(units = y_train1.shape[1], activation = 'softmax'))
         #compile, train and load CNN model
         cnn_model.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics = ['accuracy'])
         if os.path.exists("model/cnn_weights.hdf5") == False:
```

16. Conclusion

- Botnet attacks are a major threat to SDN networks.
- Traditional detection systems have limitations in accuracy and scalability.
- The proposed deep learning based system effectively detects botnet attacks.
- Among all tested algorithms, **CNN achieved the highest accuracy**.
- The system improves network security by detecting and mitigating malicious traffic.
- Future work can include **real-time deployment in large scale SDN networks**.

17. References

1.IEEE Transactions on Machine Learning – Botnet Detection in SDN.

2.Research papers on Deep Learning based Intrusion Detection Systems.

3.SDN security research articles.

4.Python Machine Learning documentation.

5.Network Security and Intrusion Detection research publications.